



TryNex Lifestyle

CACHE ARCHITECTURE REPORT

Redis strategy, MISS!'HIT flow, TTL configuration, and cache invalidation

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Primary cache

Fallback cache

in-process map (no Redis dependency in dev)

Cache status header

X-Cache-Status: HIT | MISS on all cached routes

Cache verified

X-Cache-Status: HIT confirmed on /api/products '

HTTP headers

Cache-Control: public, max-age=10, s-maxage=30, stale-while-revalidate=60

Admin requests

Cache-Control: no-store (always bypass)

Client Request: GET /api/products

Express Route Handler

! " productCacheKey() generates namespace key

redisCacheGet(key) ! ' Upstash Redis REST call

! " MISS? (first request or TTL expired)

PostgreSQL query via Drizzle ORM

! " Format response payload

redisCacheSet(key, payload, 60) ! ' Store in Redis

res.set("X-Cache-Status", "MISS") ! ' Return response

! " HIT? (subsequent request within TTL)

Return cached JSON from Redis directly

res.set("X-Cache-Status", "HIT") ! ' Return response (~2ms)

NAMESPACE	COVERS	TTL	INVALIDATION TRIGGER
	Product list (paginated slices)	60 seconds	On any product create/update/delete
trynex:categories:*	All categories	300 seconds	On category create/update/delete
trynex:settings:*	Site settings (all 80+)	60 seconds	On PUT /api/settings
trynex:blog:*	Blog posts (paginated + single)	60 seconds	On blog post create/update
trynex:public-stats:*	Today orders / total orders	30 seconds	Automatic TTL expiry
trynex:hamper:*	Gift hamper listings	120 seconds	On hamper CRUD

- All write mutations (POST/PUT/PATCH/DELETE) call invalidateXxxCache() before returning response
- Wildcard key pattern busting: common paginated slices deleted eagerly
- No Redis SCAN needed — key patterns are deterministic and pre-known
- Settings cache cleared immediately on any settings update
- Admin requests always bypass cache (Cache-Control: no-store)

- If UPSTASH_REDIS_REST_TOKEN is not set, an in-process Map is used instead
- Fallback Map provides same API surface: get(), set(), del()
- No code changes required to switch between Redis and in-process cache
- Cache entries in Map have no persistent cross-restart TTL — fully ephemeral
- Current state: Redis not configured — Map fallback is active in dev workspace

Cache HIT response time

~2–5ms (Redis REST + JSON parse)

Cache MISS response time

~10–40ms (PostgreSQL + Redis SET)

Products cache HIT confirmed

X-Cache-Status: HIT on 2nd request '

Cloudflare CDN layer

Client-side browser cache

stale-while-revalidate

Recommendation

s-maxage=30 !' Cloudflare caches at edge for 30s

max-age=10 !' browsers hold for 10 seconds

60s !' background revalidation, no user-visible wait

Set UPSTASH_REDIS_REST_TOKEN in Render env vars for production Redis